

Department of Computer Engineering - Semester 3 Syllabus

TEST DATA - NOT AN OFFICIAL SVIT DOCUMENT

Subject: Data Structures and Algorithms (Subject Code: CS-301)

Academic Year: 2026-2027 | Regulation: GTU CBCS Version 2026

Unit 1: Linear Data Structures - Arrays, Stacks, Queues, Linked Lists.

Unit 2: Non-Linear Data Structures - Binary Search Trees, AVL Trees, Heaps.

Unit 3: Graph Algorithms - BFS, DFS, Dijkstra Shortest Path, Minimum Spanning Trees.

Unit 4: Searching and Sorting - QuickSort, MergeSort, RadixSort, Hashing with Chaining.

Textbook Reference: Introduction to Algorithms by Cormen, Leiserson, Rivest, Stein.